

## Demonstration of Proposed Features

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 28 June 2026  |
| <b>Team ID</b>       | XXXXXX        |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 1 Mark        |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

| S.No | Feature Name                          | Description                                                                                                                                                  | Status<br>(Implemented /<br>Partial /<br>Pending) | Demonstrate<br>d (Yes / No) | Remarks                                                                                               |
|------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------|
| 1    | <b>ML-Based Flood Classification</b>  | Train and evaluate four classification algorithms — Decision Tree, Random Forest, KNN, and XGBoost — on historical weather data to predict flood occurrence. | <b>Implemented</b>                                | <b>Yes</b>                  | All four models trained and evaluated. XGBoost selected as best model with 96.55% accuracy.           |
| 2    | <b>Data Preprocessing Pipeline</b>    | Process raw meteorological data including feature scaling, encoding categorical variables, and train/test splitting for model training.                      | <b>Implemented</b>                                | <b>Yes</b>                  | Pipeline includes feature scaling and train/test split. Dataset prepared from historical CSV records. |
| 3    | <b>Model Persistence &amp; Saving</b> | Save the best-performing model (XGBoost) using pickle/joblib for later loading and inference in the web application.                                         | <b>Implemented</b>                                | <b>Yes</b>                  | XGBoost model saved successfully. Model loaded and used in Flask app for real-time predictions.       |

|   |                                   |                                                                                                                                              |             |     |                                                                                                 |
|---|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|-------------------------------------------------------------------------------------------------|
| 4 | Flask Web Application             | Build a web interface that allows users to input meteorological features and receive instant flood risk predictions from the saved model.    | Implemented | Yes | Flask app developed with input form and prediction output. REST API endpoints functional.       |
| 5 | Multi-Region Monitoring Dashboard | Enable disaster relief coordinators to enter weather data for multiple regions and receive simultaneous flood risk classifications.          | Implemented | Yes | Multiple regional inputs supported via the web interface. Helps prioritise resource deployment. |
| 6 | IBM Cloud Deployment              | Deploy the Flask application to IBM Cloud to make it scalable and accessible from any location by authorities and disaster management teams. | Implemented | Yes | Application configured for IBM Cloud deployment. Scalable and globally accessible via browser.  |

Feature Implementation Summary:

|                                 |      |
|---------------------------------|------|
| Total Features Proposed         | 6    |
| Total Features Implemented      | 6    |
| Total Features Demonstrated     | 6    |
| Overall Implementation Rate (%) | 100% |